SRM INSTITUTE OF SCIENCE AND TECHNOLOGY

TUTORIAL ASSIGNMENT - 2

DATA WAREHOUSING AND DATA MINING

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Batch : III Year / V Sem / BDA “C”

Subject : Data Warehousing and Data Mining

Assignment Description

This tutorial assignment was performed using Orange Data Mining, a visual data mining and machine learning tool. The practicals demonstrate data preparation, transformation, association rule mining, statistical analysis, regression, outlier analysis, and clustering through Orange widgets and visual workflows.

Experiments Included

1. Data Cleaning on Dataset using Orange
2. Data Reduction on Dataset using Orange
3. Data Transformation using Orange
4. Generate Association Rules using Apriori Algorithm
5. Statistical Analysis on Dataset using Orange
6. Performing Regression on Dataset
7. Implementation of Outlier Analysis using Orange
8. K-Means Algorithm for Iris Dataset
9. Clustering on iris.arff using Simple K-Means

Submitted as part of the Data Warehousing and Data Mining Tutorial Assignment - 27/10/20206

1: DATA CLEANING USING ORANGE

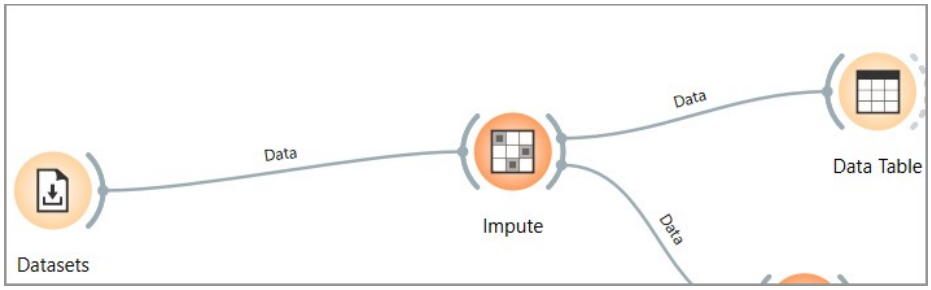
Aim: To perform data cleaning on a dataset using Orange Data Mining.

Software: Orange Data Mining. Widgets: Datasets, Impute, Data Table.

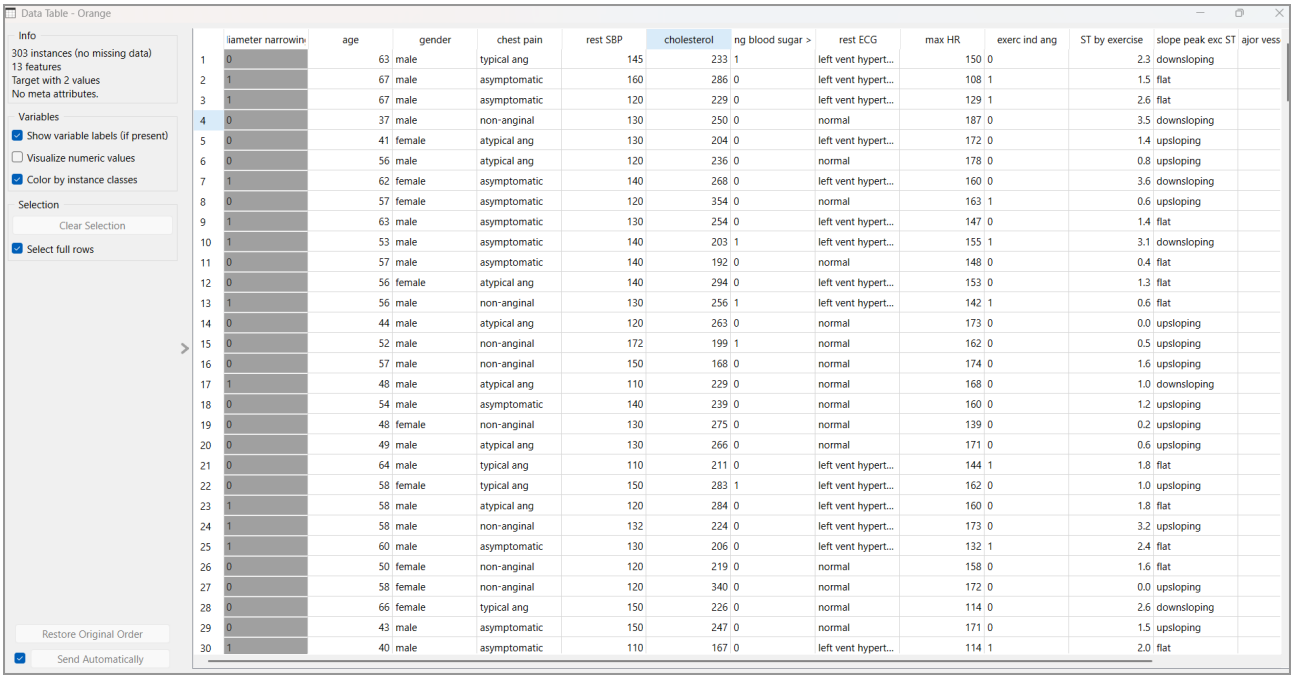
Theory: Data cleaning detects and corrects problems in raw data, such as missing values, so that later analysis is reliable. The Impute widget replaces missing values with substitutes such as the average or the most frequent value.

Procedure: Load the Heart Disease dataset (303 instances, 13 features) with the Datasets widget. Connect it to Impute to fill in the missing values, then connect Impute to a Data Table to inspect the cleaned data.

Output screenshots



Workflow: Datasets → Impute → Data Table



	iameter narrowin	age	gender	chest pain	rest SBP	cholesterol	ng blood sugar >	rest ECG	max HR	exerc ind ang	ST by exercise	slope peak exc ST	ajor vess
1	0	63	male	typical ang	145	233	1	left vent hypert...	150	0	2.3	downsloping	
2	1	67	male	asymptomatic	160	286	0	left vent hypert...	108	1	1.5	flat	
3	1	67	male	asymptomatic	120	229	0	left vent hypert...	129	1	2.6	flat	
4	0	37	male	non-anginal	130	250	0	normal	187	0	3.5	downsloping	
5	0	41	female	atypical ang	130	204	0	left vent hypert...	172	0	1.4	upsloping	
6	0	56	male	atypical ang	120	236	0	normal	178	0	0.8	upsloping	
7	1	62	female	asymptomatic	140	268	0	left vent hypert...	160	0	3.6	downsloping	
8	0	57	female	asymptomatic	120	354	0	normal	163	1	0.6	upsloping	
9	1	63	male	asymptomatic	130	254	0	left vent hypert...	147	0	1.4	flat	
10	1	53	male	asymptomatic	140	203	1	left vent hypert...	155	1	3.1	downsloping	
11	0	57	male	asymptomatic	140	192	0	normal	148	0	0.4	flat	
12	0	56	female	atypical ang	140	294	0	left vent hypert...	153	0	1.3	flat	
13	1	56	male	non-anginal	130	256	1	left vent hypert...	142	1	0.6	flat	
14	0	44	male	atypical ang	120	263	0	normal	173	0	0.0	upsloping	
15	0	52	male	non-anginal	172	199	1	normal	162	0	0.5	upsloping	
16	0	57	male	non-anginal	150	168	0	normal	174	0	1.6	upsloping	
17	1	48	male	atypical ang	110	229	0	normal	168	0	1.0	downsloping	
18	0	54	male	asymptomatic	140	239	0	normal	160	0	1.2	upsloping	
19	0	48	female	non-anginal	130	275	0	normal	139	0	0.2	upsloping	
20	0	49	male	atypical ang	130	266	0	normal	171	0	0.6	upsloping	
21	0	64	male	typical ang	110	211	0	left vent hypert...	144	1	1.8	flat	
22	0	58	female	typical ang	150	283	1	left vent hypert...	162	0	1.0	upsloping	
23	1	58	male	atypical ang	120	284	0	left vent hypert...	160	0	1.8	flat	
24	1	58	male	non-anginal	132	224	0	left vent hypert...	173	0	3.2	upsloping	
25	1	60	male	asymptomatic	130	206	0	left vent hypert...	132	1	2.4	flat	
26	0	50	female	non-anginal	120	219	0	normal	158	0	1.6	flat	
27	0	58	female	non-anginal	120	340	0	normal	172	0	0.0	upsloping	
28	0	66	female	typical ang	150	226	0	normal	114	0	2.6	downsloping	
29	0	43	male	asymptomatic	150	247	0	normal	171	0	1.5	upsloping	
30	1	40	male	asymptomatic	110	167	0	left vent hypert...	114	1	2.0	flat	

Cleaned dataset in the Data Table (303 instances, no missing data)

Result: The Data Table reports 303 instances with no missing data, confirming that the missing values were imputed.

2: DATA REDUCTION ON DATASET USING ORANGE

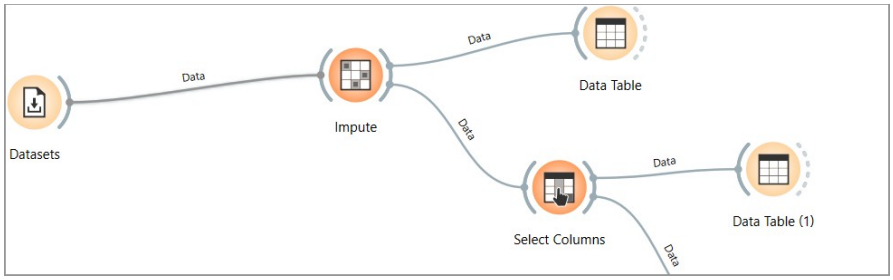
Aim: To perform data reduction on a dataset using Orange Data Mining by removing unnecessary attributes.

Software: Orange Data Mining. Widgets: Datasets, Impute, Select Columns, Data Table.

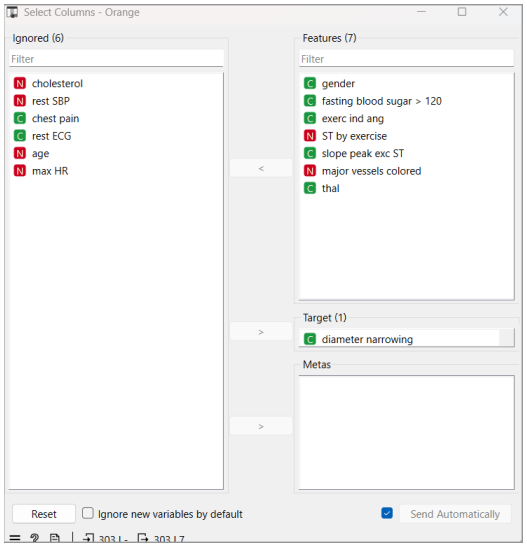
Theory: Data reduction lowers the number of attributes while keeping the information needed for analysis, which makes the data easier to store and process.

Procedure: Connect the imputed data to Select Columns. Move cholesterol, rest SBP, chest pain, rest ECG, age and max HR to Ignored, keep seven features, and set “diameter narrowing” as the Target. Connect the output to Data Table (1).

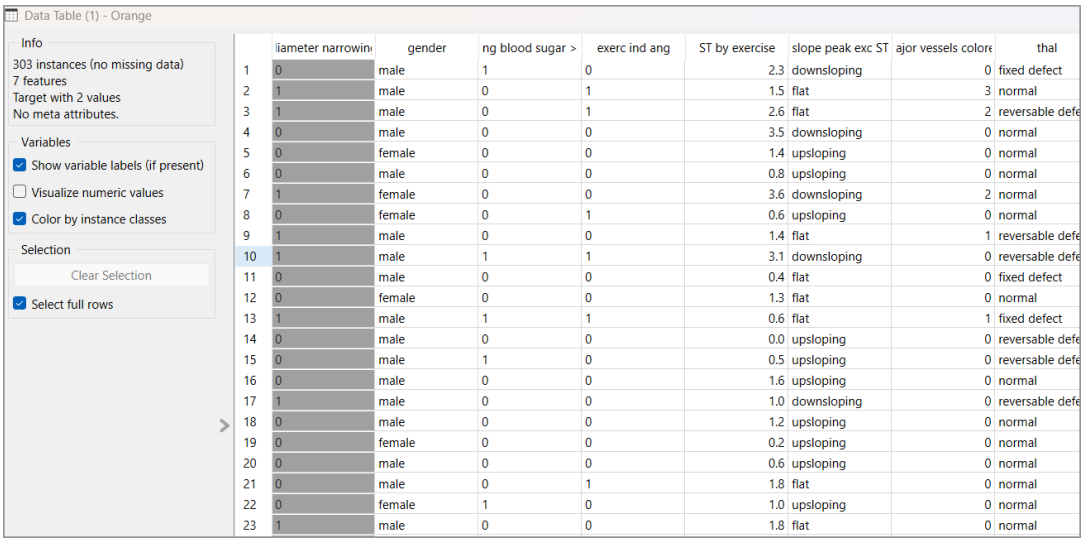
Output screenshots



Workflow: Impute → Select Columns → Data Table (1)



Select Columns function



	iameter narrowin	gender	ng blood sugar >	exerc ind ang	ST by exercise	slope peak exc ST	ajor vessels colore	thal
1	0	male	1	0	2.3	downsloping	0	fixed defect
2	1	male	0	1	1.5	flat	3	normal
3	1	male	0	1	2.6	flat	2	reversable defe
4	0	male	0	0	3.5	downsloping	0	normal
5	0	female	0	0	1.4	upsloping	0	normal
6	0	male	0	0	0.8	upsloping	0	normal
7	1	female	0	0	3.6	downsloping	2	normal
8	0	female	0	1	0.6	upsloping	0	normal
9	1	male	0	0	1.4	flat	1	reversable defe
10	1	male	1	1	3.1	downsloping	0	reversable defe
11	0	male	0	0	0.4	flat	0	fixed defect
12	0	female	0	0	1.3	flat	0	normal
13	1	male	1	1	0.6	flat	1	fixed defect
14	0	male	0	0	0.0	upsloping	0	reversable defe
15	0	male	1	0	0.5	upsloping	0	reversable defe
16	0	male	0	0	1.6	upsloping	0	normal
17	1	male	0	0	1.0	downsloping	0	reversable defe
18	0	male	0	0	1.2	upsloping	0	normal
19	0	female	0	0	0.2	upsloping	0	normal
20	0	male	0	0	0.6	upsloping	0	normal
21	0	male	0	1	1.8	flat	0	normal
22	0	female	1	0	1.0	upsloping	0	normal
23	1	male	0	0	1.8	flat	0	normal

Data after reduction (7 features)

Result: The dataset was reduced from 13 features to 7 features for the same 303 instances.

3: DATA TRANSFORMATION USING ORANGE

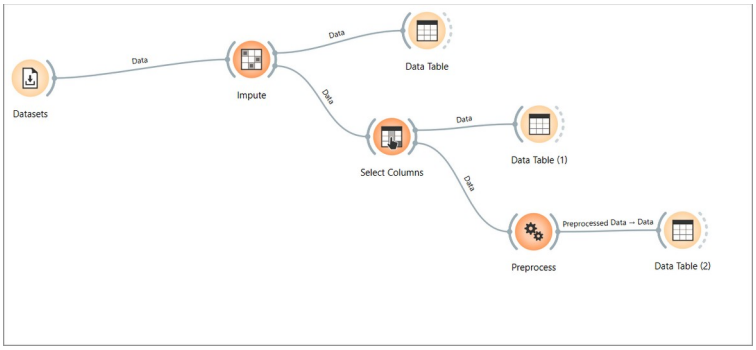
Aim: To perform data transformation on a dataset using Orange Data Mining by standardizing the numeric attributes of the selected columns.

Software: Orange Data Mining. Widgets: Datasets, Impute, Select Columns, Preprocess, Data Table.

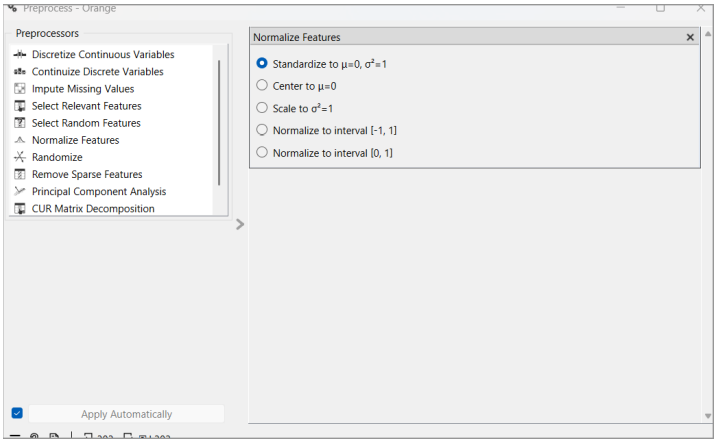
Theory: Data transformation converts data into a form suitable for mining. Standardization rescales a numeric attribute to mean 0 and variance 1, so that attributes measured on different scales become comparable.

Procedure: Connect Select Columns to the Preprocess widget and add the Normalize Features preprocessor with the option “Standardize to μ=0, σ²=1”. Connect the output to Data Table (2).

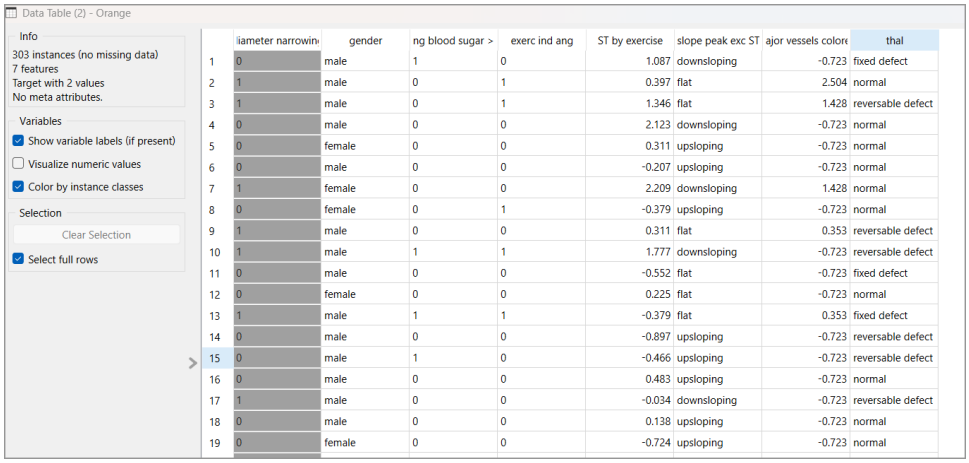
Output screenshots



Steps taken: Select Columns → Preprocess → Data Table (2)



Preprocess widget: Normalize Features



	iameter narrowin	gender	ng blood sugar >	exerc ind ang	ST by exercise	slope peak exc ST	ajor vessels colore	thal
1	0	male	1	0	1.087	downsloping	-0.723	fixed defect
2	1	male	0	1	0.397	flat	2.504	normal
3	1	male	0	1	1.346	flat	1.428	reversable defect
4	0	male	0	0	2.123	downsloping	-0.723	normal
5	0	female	0	0	0.311	upsloping	-0.723	normal
6	0	male	0	0	-0.207	upsloping	-0.723	normal
7	1	female	0	0	2.209	downsloping	1.428	normal
8	0	female	0	1	-0.379	upsloping	-0.723	normal
9	1	male	0	0	0.311	flat	0.353	reversable defect
10	1	male	1	1	1.777	downsloping	-0.723	reversable defect
11	0	male	0	0	-0.552	flat	-0.723	fixed defect
12	0	female	0	0	0.225	flat	-0.723	normal
13	1	male	1	1	-0.379	flat	0.353	fixed defect
14	0	male	0	0	-0.897	upsloping	-0.723	reversable defect
15	0	male	1	0	-0.466	upsloping	-0.723	reversable defect
16	0	male	0	0	0.483	upsloping	-0.723	normal
17	1	male	0	0	-0.034	downsloping	-0.723	reversable defect
18	0	male	0	0	0.138	upsloping	-0.723	normal
19	0	female	0	0	-0.724	upsloping	-0.723	normal

New transformed data

Result: The numeric attributes (ST by exercise, major vessels colored) now hold standardized values such as 1.087 and −0.723, while the 303 instances and 7 features are retained.

4: GENERATE ASSOCIATION RULES USING APRIORI ALGORITHM IN ORANGE

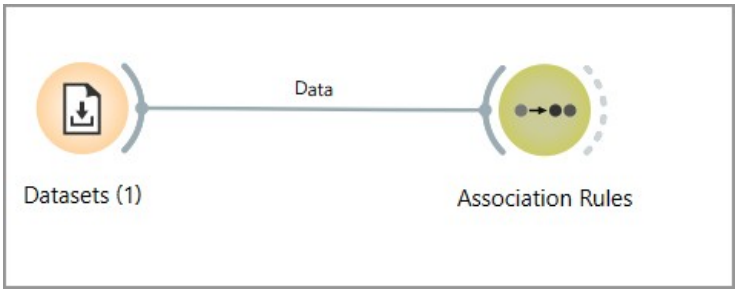
Aim: To generate association rules from a transaction dataset using the Apriori algorithm in Orange Data Mining.

Software: Orange Data Mining. Widgets: Datasets, Association Rules.

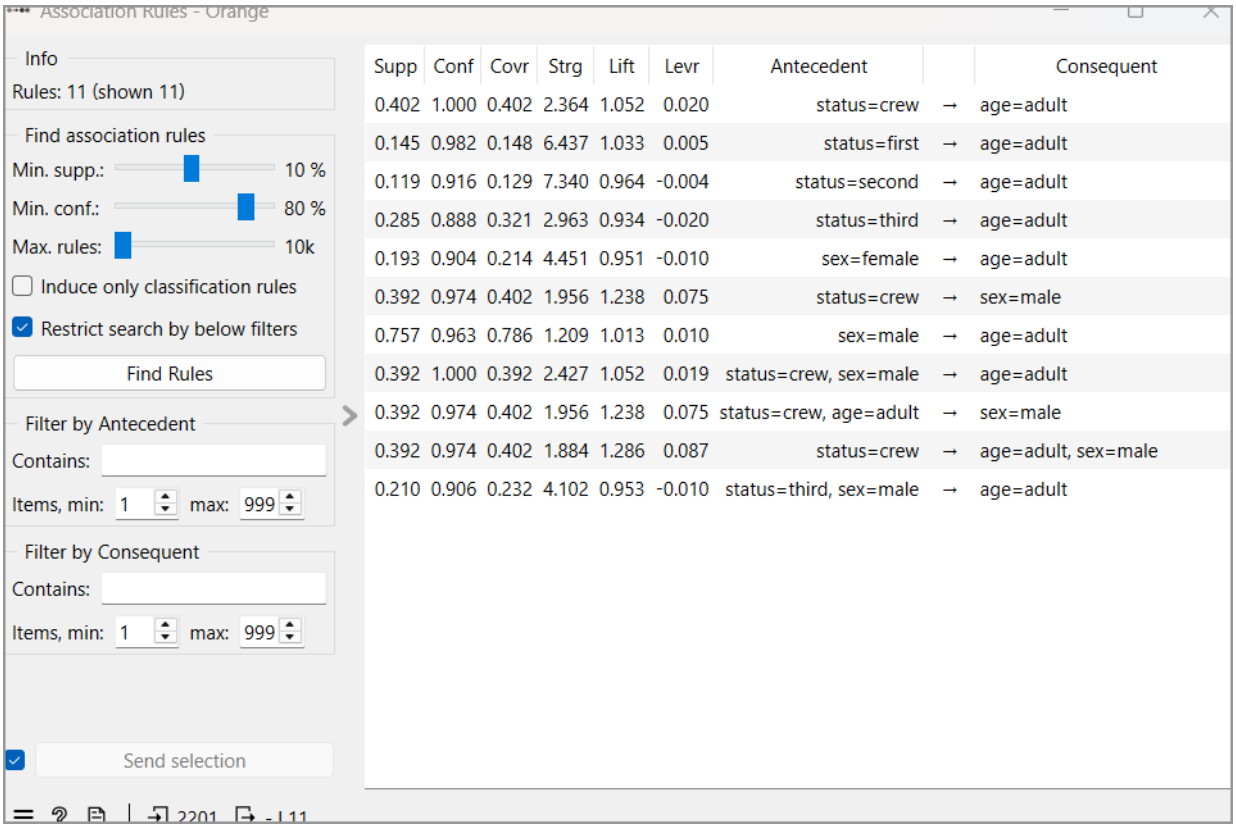
Theory: Association rule mining finds items or attribute values that occur together. A rule A → B is judged by its support (how often A and B occur together), confidence (how often B occurs when A does) and lift (how much more often they occur together than by chance).

Procedure: Load the Titanic dataset (2201 instances) with the Datasets widget and connect it to Association Rules. Set Min. support to 10% and Min. confidence to 80%, then click Find Rules.

Output screenshots



Steps: Datasets → Association Rules



Association rules generated (11 rules)

Result: 11 rules were found. For example, status=crew → age=adult has support 0.402, confidence 1.000 and lift 1.052, and status=crew, sex=male → age=adult also has confidence 1.000.

5: STATISTICAL ANALYSIS ON DATASET USING ORANGE

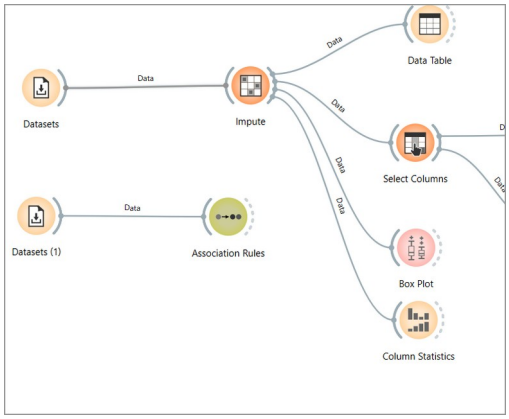
Aim: To perform statistical analysis on a dataset using Orange Data Mining.

Software: Orange Data Mining. Widgets: Datasets, Impute, Column Statistics, Box Plot.

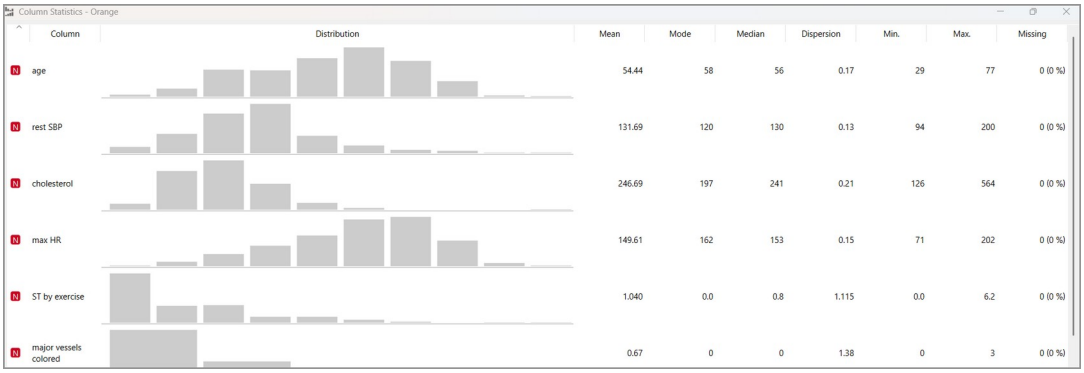
Theory: Statistical analysis summarizes attributes and displays distributions, spread and summary measures such as mean, median, mode, minimum and maximum.

Procedure: Connect the imputed Heart Disease data to Column Statistics and to Box Plot. Inspect the numeric columns in Column Statistics and view age grouped by “diameter narrowing” in the Box Plot.

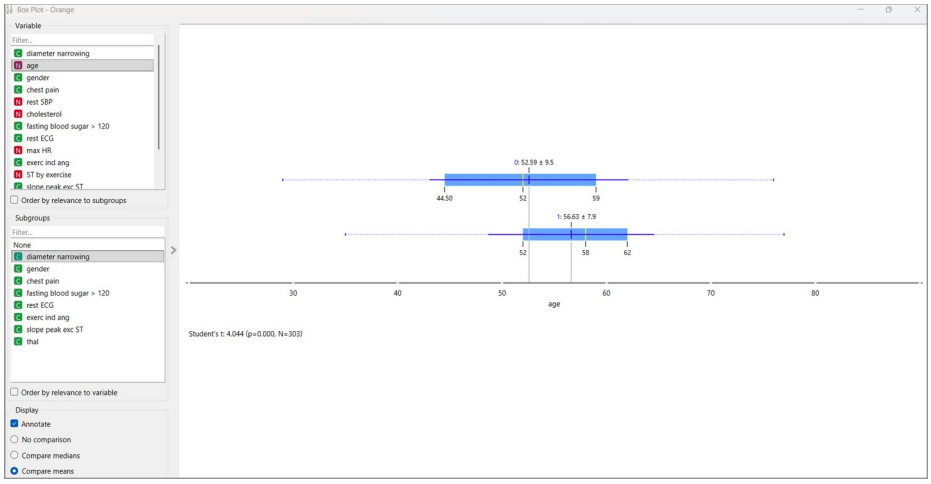
Output screenshots



Workflow: Impute → Column Statistics and Box Plot



Column Statistics



Box Plot: age by diameter narrowing

Result: Column Statistics shows, for example, age with mean 54.44 and median 56 (range 29–77), cholesterol with mean 246.69 and median 241 (range 126–564), and max HR with mean 149.61 (range 71–202). The Box Plot shows a mean age of 52.59 for class 0 against 56.63 for class 1 (Student’s t = 4.044, p = 0.000).

6: PERFORMING REGRESSION USING ORANGE

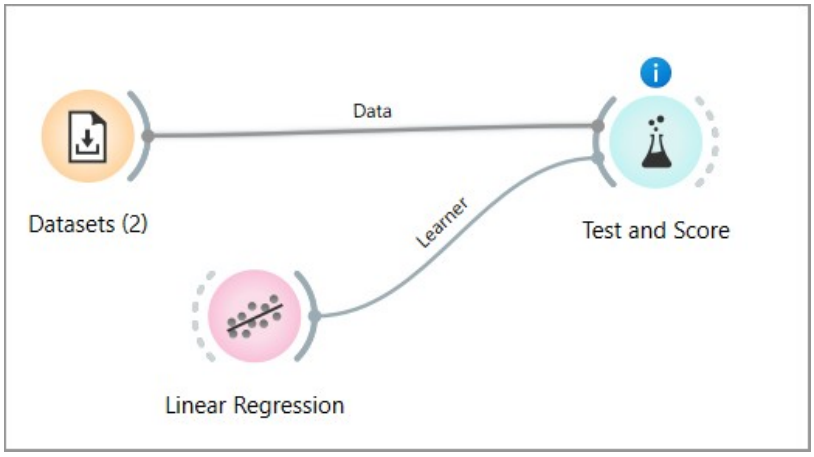
Aim: To perform regression on a dataset using Orange Data Mining.

Software: Orange Data Mining. Widgets: Datasets, Linear Regression, Test and Score.

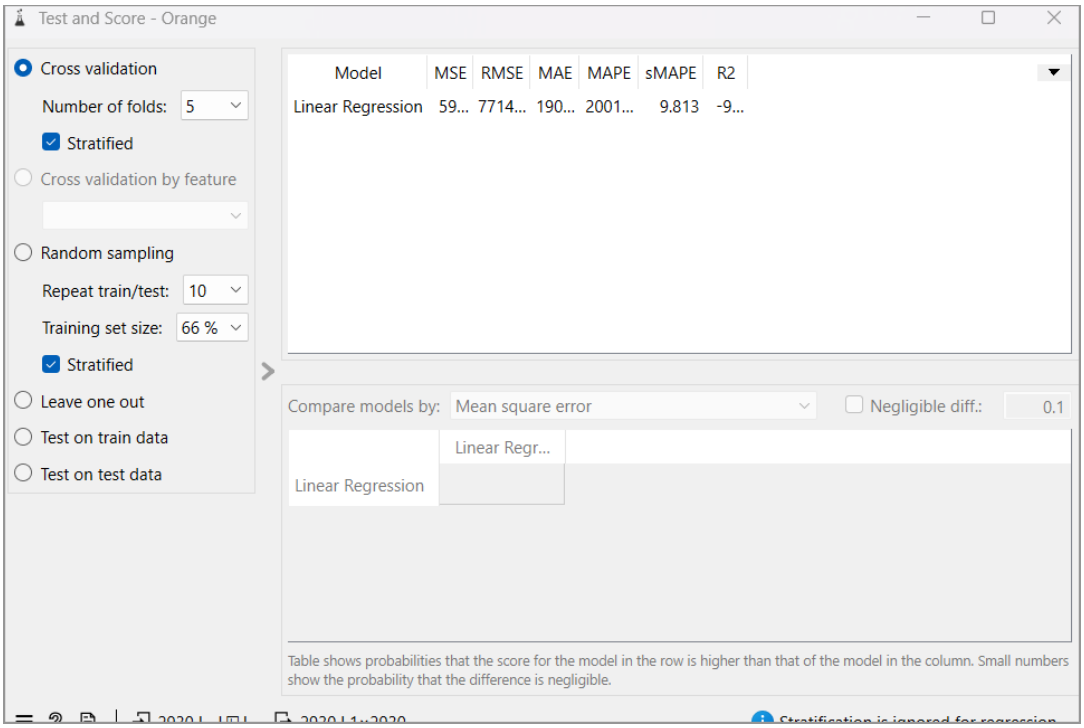
Theory: Regression predicts a numerical target from input features. Linear Regression fits a linear model, and Test and Score evaluates it with measures such as MSE, RMSE, MAE and R^2.

Procedure: Load a dataset with a numeric target (2930 instances) using the Datasets widget and connect it to Test and Score. Connect the Linear Regression learner to the Learner input and evaluate with 5-fold cross validation.

Output screenshots



Workflow: Datasets → Test and Score ← Linear Regression



Test and Score results

Result: Test and Score evaluated the Linear Regression model with 5-fold cross validation and reported MSE, RMSE, MAE, MAPE, sMAPE and R^2.

7: IMPLEMENTATION OF OUTLIER ANALYSIS USING ORANGE

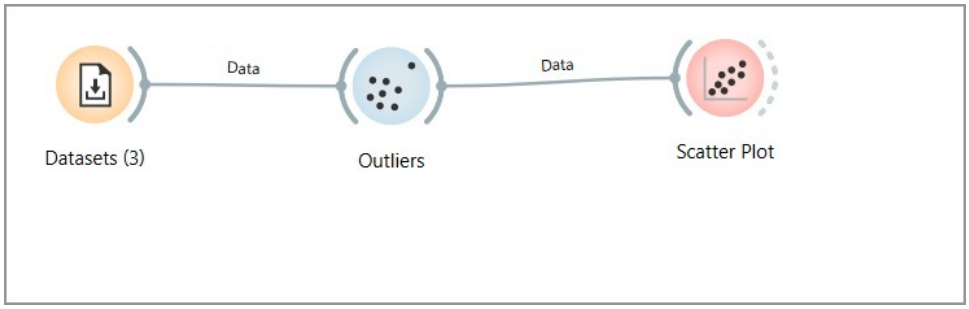
Aim: To identify potential outliers in a dataset using Orange Data Mining.

Software: Orange Data Mining. Widgets: Datasets, Outliers, Scatter Plot.

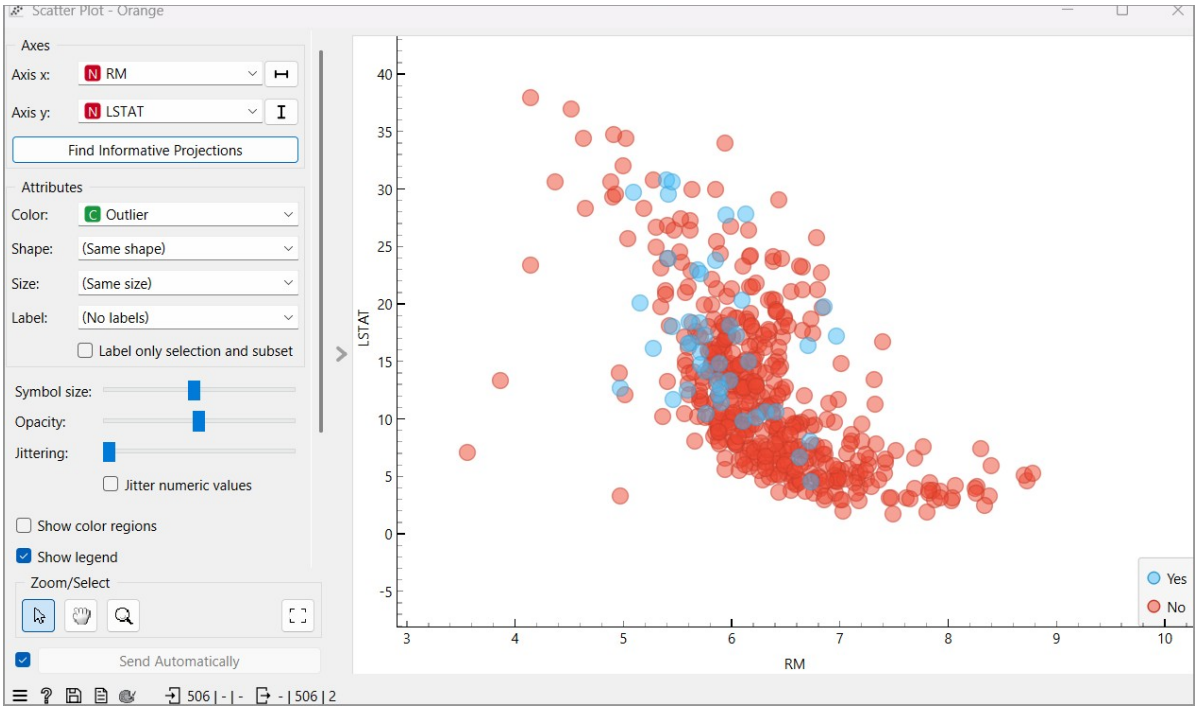
Theory: Outlier analysis identifies observations that differ from the general pattern of the data and should be examined. The Outliers widget marks each instance as an outlier (Yes) or not (No).

Procedure: Load the housing dataset (506 instances) with the Datasets widget and connect it to Outliers, then to Scatter Plot. Plot RM on the x-axis against LSTAT on the y-axis and colour the points by Outlier.

Output screenshots



Workflow: Datasets → Outliers → Scatter Plot



Scatter plot coloured by Outlier (Yes = blue, No = red)

Result: The Outliers widget flagged unusual instances, which the scatter plot shows in blue (Yes) among the regular instances in red (No).

8: IMPLEMENTATION OF K-MEANS ALGORITHM FOR IRIS DATASET

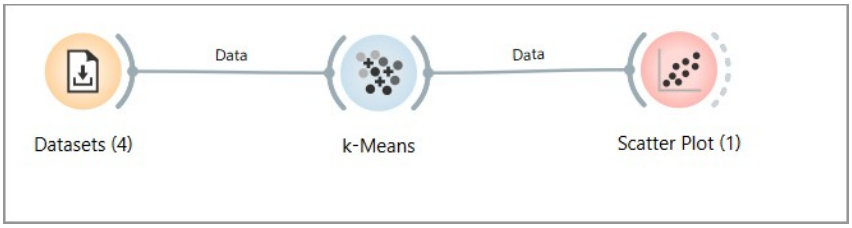
Aim: To implement K-Means clustering on the Iris dataset using Orange Data Mining.

Software: Orange Data Mining. Widgets: Datasets, k-Means, Scatter Plot.

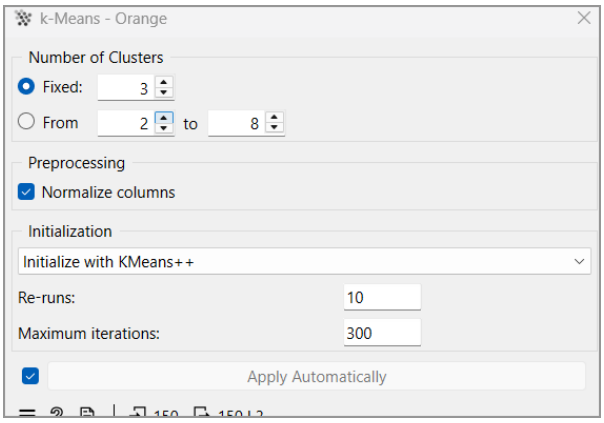
Theory: K-Means is an unsupervised algorithm that groups observations into K clusters based on feature similarity.

Procedure: Load Iris (150 instances) with the Datasets widget and connect it to k-Means. Set a fixed number of 3 clusters with Normalize columns on and KMeans++ initialization. Connect the output to Scatter Plot (1), choose petal length and petal width, and colour by Cluster.

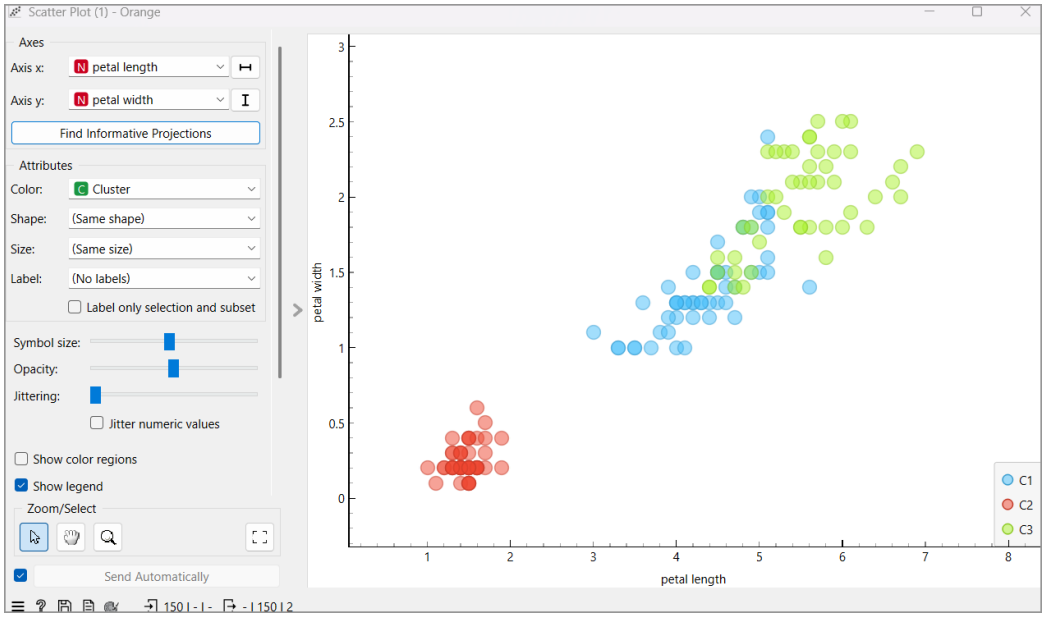
Output screenshots



Workflow: Datasets → k-Means → Scatter Plot (1)



k-Means settings (K = 3)



Iris K-Means clusters: petal length vs petal width

Result: K-Means clustering with K=3 split Iris into clusters C1, C2 and C3. C2 is clearly separated, while C1 and C3 sit next to each other with slight overlap.

9: DEMONSTRATION OF CLUSTERING PROCESS ON iris.arff USING SIMPLE K-MEANS

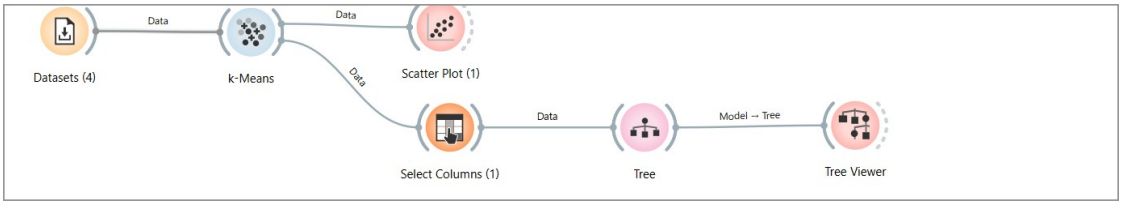
Aim: To demonstrate clustering on the iris data using K-Means in Orange Data Mining and to interpret the resulting clusters.

Software: Orange Data Mining. Widgets: Datasets, k-Means, Select Columns, Tree, Tree Viewer.

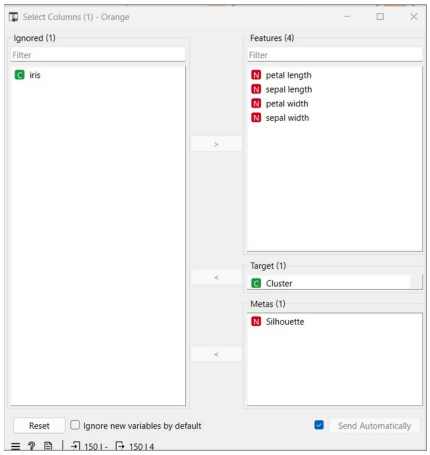
Theory: Clustering groups similar observations without using class labels. A decision tree trained on the cluster labels shows which attributes separate the clusters.

Procedure: Run k-Means with K=3 on Iris. Connect the output to Select Columns (1), keep the four measurements as features, set Cluster as Target and ignore the original iris class. Connect it to Tree and then to Tree Viewer.

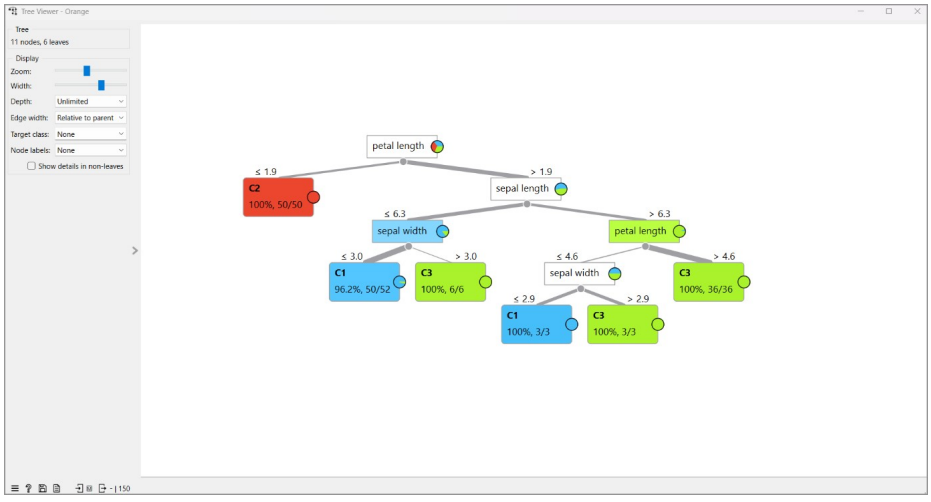
Output screenshots



Workflow: k-Means → Select Columns (1) → Tree → Tree Viewer



Select Columns (1): Cluster as target

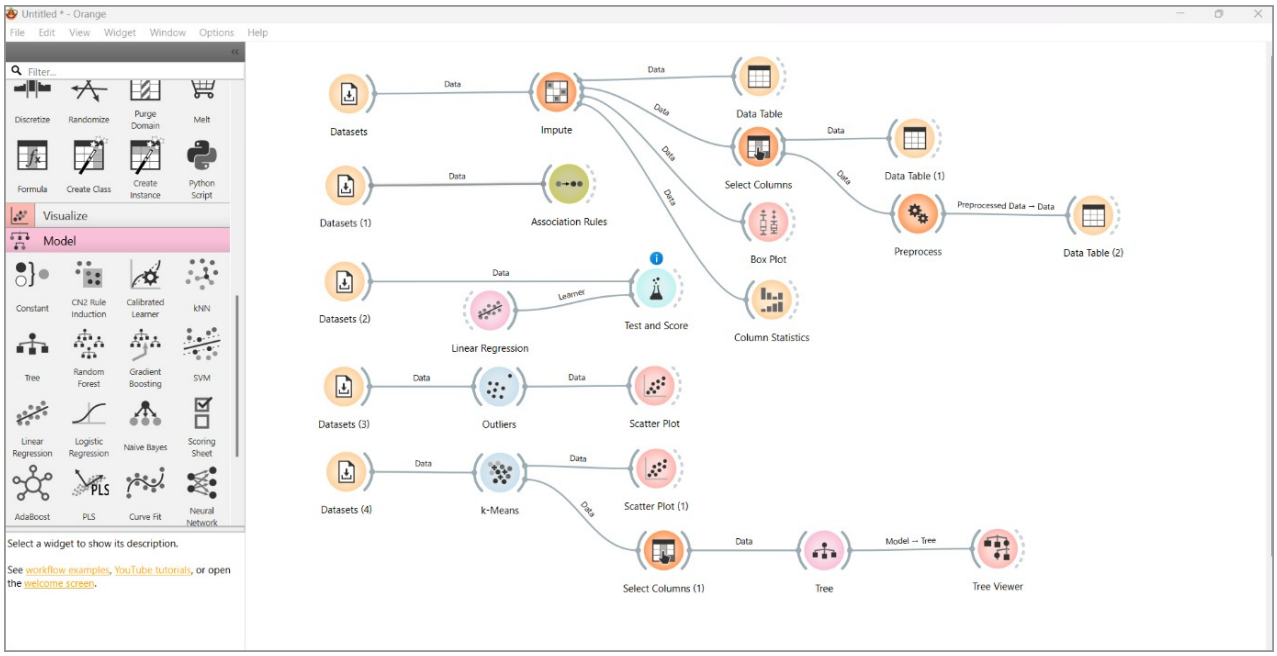


Tree Viewer: tree built on the cluster labels

Result: Clustering was demonstrated with K=3. The tree (11 nodes, 6 leaves) first splits on petal length ≤ 1.9, which isolates cluster C2 completely (50 of 50), and then separates C1 and C3 using sepal length, sepal width and petal width.

COMPLETE ORANGE WORKFLOW CANVAS

All the workflows from experiments 1 to 9 as they appear together on a single Orange canvas.



Orange canvas containing the workflows of all experiments